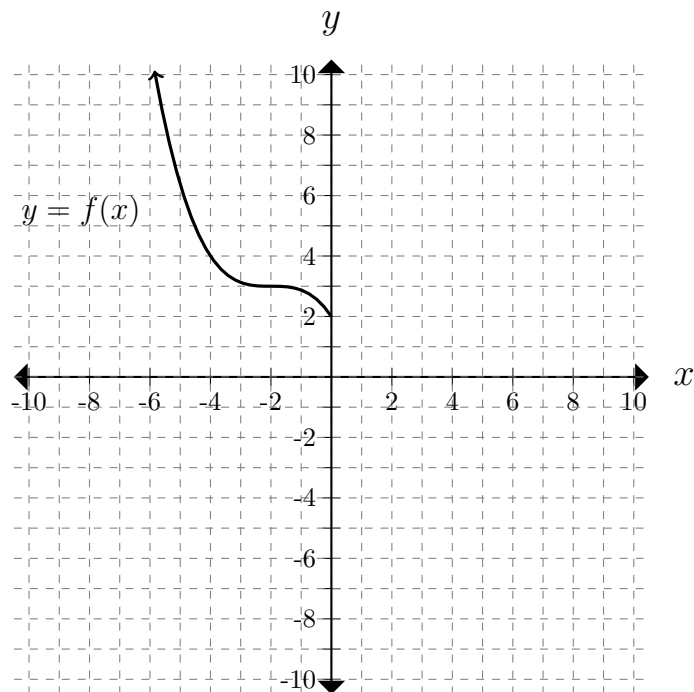


Workshop 2

Problem 1.

Keywords: Inverse Functions, Horizontal Line Test, Domain, Range

Consider the function f whose graph is sketched below.



- (a) The function f is one-to-one. Explain why in one or two sentences.
- (b) Find the domain and range of f and write each (carefully labeled) using interval notation.
- (c) Find the domain and range of f^{-1} and write each (carefully labeled) using interval notation.
- (d) Find the value of $f^{-1}(2)$ and $f^{-1}(4)$.
- (e) Make a careful sketch of the graph of $y = f^{-1}(x)$ on the set of axes above. Be sure to label your graph appropriately.

Problem 2.

Keywords: Distance, Equations of Circles, Vertical Line Test

- (a) Consider the point $P(-1, 2)$. Give a formula for the distance between P and a point (x, y) of the plane.

- (b) Consider the circle C of center P and radius 2. Find an equation of C .
- (c) Draw a sketch of C in the (x, y) -plane. Can C be the graph of a function? Justify your answer carefully with one or two sentences.
- (d) Try to solve the equation of C for y . How many solutions do you obtain? Explain why this is consistent with your answer to the previous question.
- (e) Find an equation of the form $y = f(x)$ for the top half semi-circle of C .
- (f) Consider the circle C' of equation $x^2 - 4x + y^2 - 6y - 3 = 0$. Find the center and radius of C' , and an equation of the form $x = g(y)$ for the bottom half semi-circle of C' .

Problem 3.

Keywords: Exponential Decay, Inverse Functions, Horizontal Line Test, Domain, Range

Problem: Cesium-137 has a half life of 30 years. Suppose you have a sample that contains 100 mg of Cesium-137.

- (a) Find a function $P(t)$ that measures the mass of the Cesium-137 left in the sample t years later. Give the domain and range for P .
- (b) Find the inverse function $P^{-1}(m)$. Be sure to give the domain and range for P^{-1} .
- (c) Using 1-2 English sentences, explain the meaning of the inverse function P^{-1} you found in part (b).
- (d) How long will it take for the mass of Cesium-137 in the sample to reach 1 mg?